



AZURE DATA ENGINEERING

40 DAYS • 1 HOUR / DAY

Design the data systems behind better decisions.

A focused Azure Data Engineering program covering the platform, the pipeline, and the analytical layer—from SQL and data lakes to Data Factory, Databricks, Synapse, and Power BI.

RPA Vault Learning Series

Azure • SQL • Data Factory • Databricks • Synapse • Power BI

[Talk to a Course Advisor →](#)

[Explore RPA Vault →](#)

THE COMPLETE LEARNING SEQUENCE

Follow the data, from source to insight

The curriculum is organized around how modern data systems are planned, built, operated, and analyzed. Each phase brings together platform concepts and applied labs.

01 • FOUNDATIONS

Model the data

Build data-warehouse and SQL fluency before working in the cloud.

02 • DATA PLATFORM

Store & govern it

Learn the core services that host, protect, and organize enterprise data.

03 • ORCHESTRATION

Move & transform it

Design Data Factory pipelines with activities, triggers, runtime options, and production controls.

04 • ANALYTICS

Scale & analyze it

Complete the platform with Databricks, Synapse Analytics, Power BI, and career readiness.

Model

Warehouse architecture, SQL, and business rules.



Store

Azure SQL, Storage, secure access, and Data Lake Gen2.



Orchestrate

ADF, Databricks, activities, triggers, and production control.



Analyze

Synapse, Serverless SQL, Power BI, and career readiness.

Applied by design: The learning path combines labs, quizzes, interface tours, and realistic data scenarios. Students learn how individual Azure services connect into a dependable end-to-end data platform.

01 • DATA WAREHOUSING, SQL & CLOUD

Start with the systems that make data useful

Before a pipeline can move data, teams need to understand what data is for, how it is modeled, and where each platform component fits.

Data warehouse fundamentals

- Data warehouse introduction, definitions, architecture, and schemas.
- Facts and dimensions: identify the measures and descriptive business context that shape analytical models.
- OLTP vs. OLAP: compare transaction processing with analytical workloads.

SQL essentials for data work

- SQL introduction and installation.
- DDL, DML, and DRL command families.
- Inner, left outer, right outer, and full outer joins.
- Select, Case, Switch, and If statements for data selection and business-rule logic.

CLOUD

Service models

Understand IaaS, PaaS, SaaS, and serverless, then explore the Azure Management Portal.

ROLE

Data engineering practice

Review database-engineering responsibilities, DevOps support for data automation, and the broader data-engineering process.

SERVICES

Azure data landscape

Survey relational and NoSQL databases, storage, ETL, Big Data, and Stream Analytics services.

Foundation outcome: Students can explain how a warehouse model, SQL logic, cloud service choice, and engineering process fit together before they begin building Azure data solutions.

02 • DATABASE, STORAGE & DATA LAKE

Build the secure foundation beneath every pipeline

This module brings together the database, storage, security, and recovery capabilities that support reliable cloud data workloads.

SQL**Azure SQL Database**

- Hosting options for SQL Server workloads in Azure.
- Create Logical SQL Server and SQL Database.
- Compare DTU and vCore tiers; scale capacity up and down.
- Point-in-time recovery and long-term backup retention.
- Geo-replication for disaster recovery.

ST**Azure Storage**

- Create a storage account.
- Create containers and blobs.
- Install and explore Azure Storage Explorer.
- Create SAS tokens and understand scoped, time-bound access.

DL**Data Lake Gen2**

- Azure Data Lake overview and architecture.
- Create a Data Lake Store Gen2 through the Portal.
- Manage data with Data Lake Store Gen2.
- Position lake storage for downstream integration and analytics.

SECURITY & ACCESS**Protected data access**

Configure firewall rules to whitelist required IP addresses at server and database level. Manage sensitive data through Dynamic Data Masking and encryption—then connect storage and data-lake patterns to controlled, scalable access.

Platform deliverable: Students can provision an Azure SQL Database, choose a capacity model, plan recovery and geo-replication, configure controlled access, protect sensitive data, create blob storage with scoped SAS access, and organize data in Data Lake Gen2 for downstream engineering workloads.

03 • ORCHESTRATION FOUNDATIONS

Turn disconnected data into a controlled pipeline

Azure Data Factory becomes the operating layer: it connects source and target systems, coordinates activity execution, and makes pipeline status observable.

STAGE 01**Define**

Create an ADF instance and tour the Pipeline, Data Flow, Monitor, Debug, Trigger, and Management Hub interfaces.

- Pipelines
- Linked Services
- Datasets
- Integration Runtime: Azure, Self-Hosted & SSIS

STAGE 02**Execute**

Build the first pipeline for a fictional company's data, then copy data from a Storage Account to Azure SQL Database.

- Lookup & Stored Procedure
- Filter & Get Metadata
- ForEach & Set Variable
- If Condition & Fail

STAGE 03**Extend**

Connect related services and reusable logic to make the pipeline responsive and scalable.

- Logic Apps + Outlook email
- Web Activity validation
- Parameters
- Execute Pipeline / nested pipeline

Lab flow

Create the service, build a first pipeline, explore the user experience, model a company's data need, and move data from cloud storage to Azure SQL Database.

Control flow

Filter records, retrieve storage blobs, iterate through stored-procedure results, assign variables, branch on a flag, and intentionally raise a controlled pipeline failure.

04 • SCALE, TRANSFORM & DEPLOY

Move from a working pipeline to a production-grade one

Students extend Data Factory into the patterns required for dependable operations, then introduce Databricks as scalable transformation compute.

DATA FACTORY IN PRODUCTION

Trigger, transform, operate & promote.

Trigger the right workload, transform it correctly, see what happened, and move it safely between environments.

Schedule

Use scheduled jobs, Tumbling Window vs standard Schedule triggers, and blob-creation events.

Transform

Mapping Data Flows & Power Query: remove nulls, handle error rows, and shape data.

Operate

Monitor pipeline behavior, optimize performance, and multi-file ingestion into Azure SQL.

Promote

Prepare Data Factory pipelines for multiple environments for repeatable deployments.

Compute (Databricks)

Create Azure Databricks, launch a Spark cluster, transform data using Scala, and develop ETL notebooks.

Orchestrate

Parameterize Databricks pipelines and invoke them directly from Azure Data Factory.

LAB PROGRESSION:

Event-driven pipeline execution → Data Flow transformations → error handling → multi-file ingestion → performance improvement → environment promotion → Databricks ETL integration.

05 • ANALYZE, REPORT & EXPLAIN YOUR WORK

Bring the data platform together with Azure Synapse Analytics

The final technical module connects Lakehouse concepts, serverless analysis, pipeline orchestration, business reporting, and career preparation.

01 • SYNAPSE FOUNDATION

Understand the workspace

- Explore traditional analytics vs Data Lakehouse concepts.
- Dedicated, Serverless & Spark pools.
- Synapse Workspace lab & RBAC permissions.
- Control Node, Compute Node, DMS & sharding patterns.

02 • SERVERLESS SQL

Analyze the lake

- Overview of SQL Serverless Pools.
- Database and External Tables lab.
- Explore & analyze data with Serverless SQL.
- Build Power BI reports connected to Serverless SQL & Data Lakes.

03 • SYNAPSE PIPELINES

Transform & monitor

- Overview of Synapse Pipelines and components.
- Transform data with Mapping Data Flows.
- Orchestrate, run & monitor pipelines.
- Quizzes reinforce each technical stage.

04 • POWER BI & CAREER

Make the work visible

- Power BI basics, DB/Excel connections & publishing.
- CV preparation and sample role explanation.
- Interview-question preparation.
- Explain an end-to-end project with confidence.

INSIGHT ROUTE

Query the lake

Use serverless SQL, external tables, and Data Lake data to investigate information without first moving it into a traditional database.

Make it useful

Build and publish Power BI reports, then use project stories, CV preparation, and interview practice to communicate the work.

Final outcome: Students can describe how data is stored, moved, transformed, analyzed, and reported across Azure—and explain the role they played in that solution during CV and interview discussions.



RPA Vault

YOUR NEXT DATA ENGINEERING MILESTONE

Turn your learning into an Azure data-engineering path.

Explore the RPA Vault learning ecosystem or speak with a course advisor about your technical background, goals, and the best next step.

Ready to start building?

Get course guidance, understand the learning pathway, and move closer to practical Azure Data Engineering capability.

[Explore RPA Vault →](#)

[Contact a Course Advisor →](#)